/ 校 级 研 究 生 创 新 项 目 /

路堑风吹雪灾害的形成与防治研究

The Formation and Prevention Study of Wind-induced Snow Disaster in Road Cuttings

汇报人/李赛	指导教师/马文勇 教授

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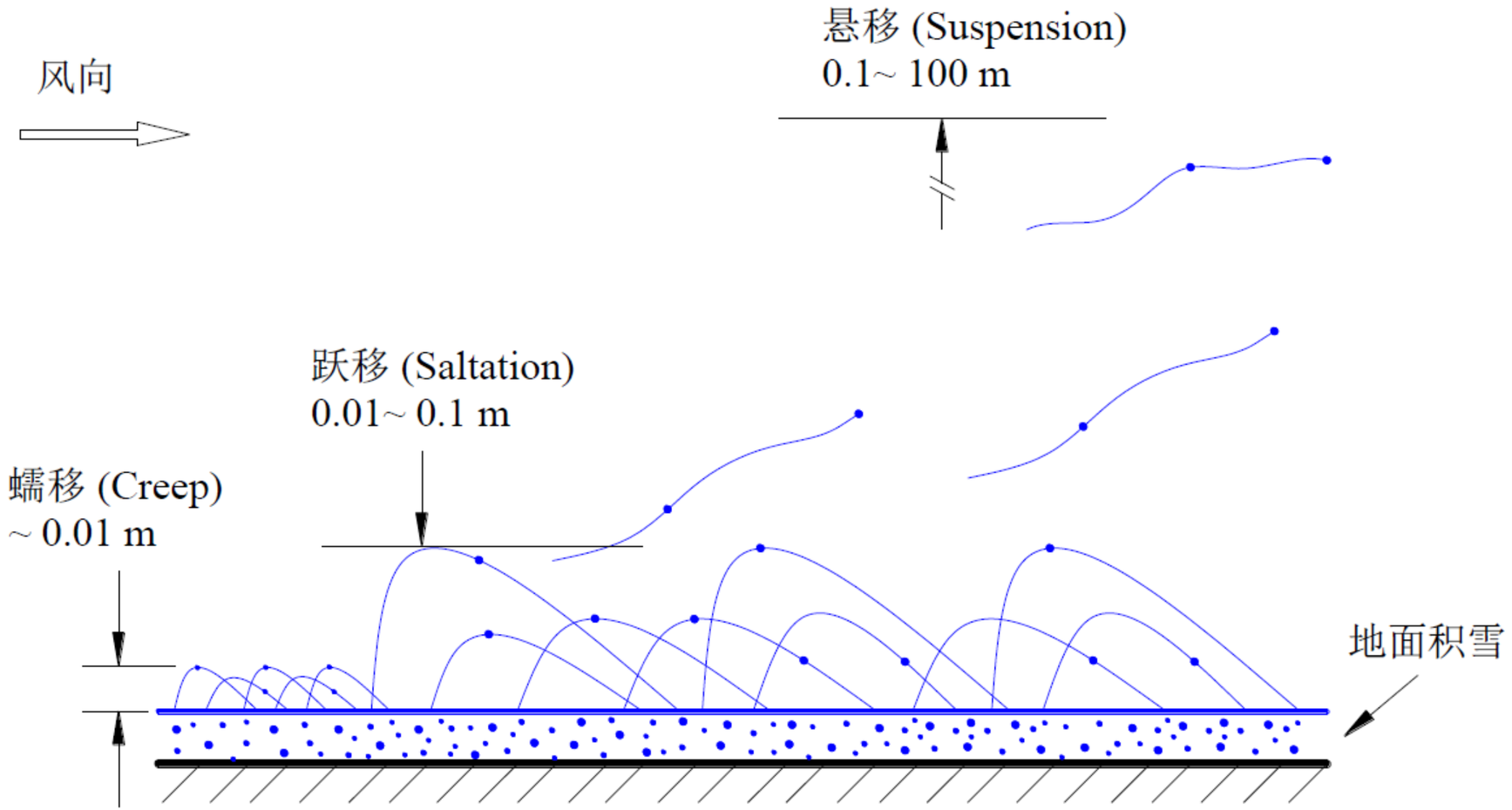
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Research Method

3 主要结论
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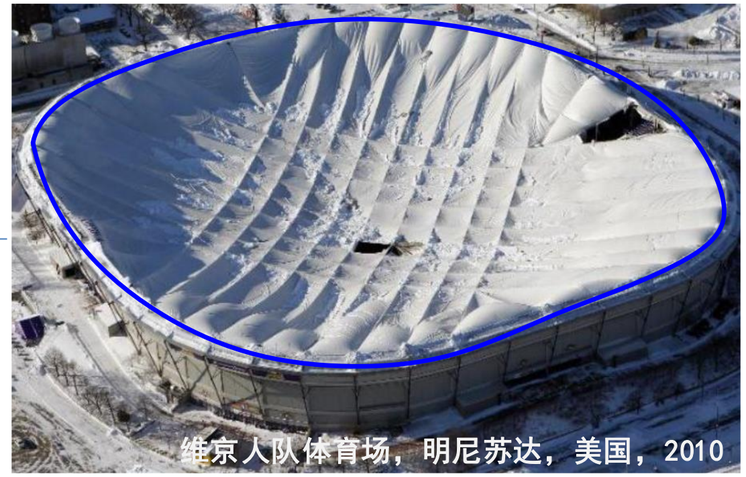
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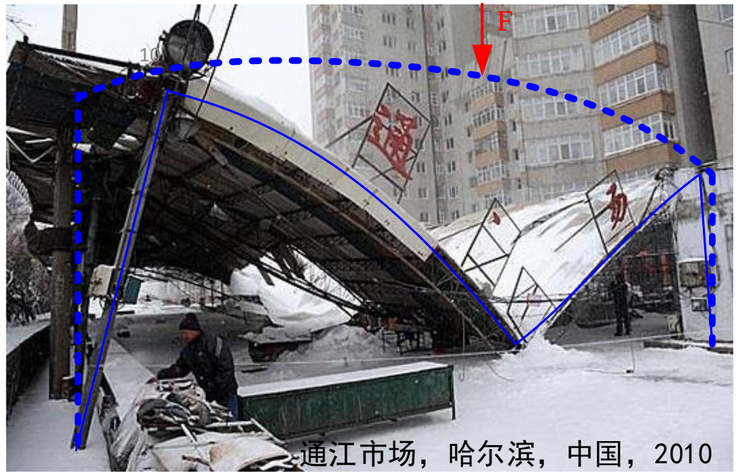
/ 风 吹 雪 现 象 /


风向
悬移 (Suspension)
0.1~ 100 m
跃移 (Saltation)
0.01~ 0.1 m
蠕移 (Creep)
~ 0.01 m
地面积雪

/ 风 吹 雪 灾 害 /

建筑屋盖


维京人队体育场，明尼苏达，美国，2010



通江市场，哈尔滨，中国，2010

交通道路


玛依塔斯地区，新疆，中国，2019


玛依塔斯地区，新疆，中国，2019

/道路风吹雪灾害/



巴州，新疆，2019



黑河，黑龙江，2019



额敏，新疆，2019



玉树，青海，2019

001 分布范围更广

002 灾害治理更复杂

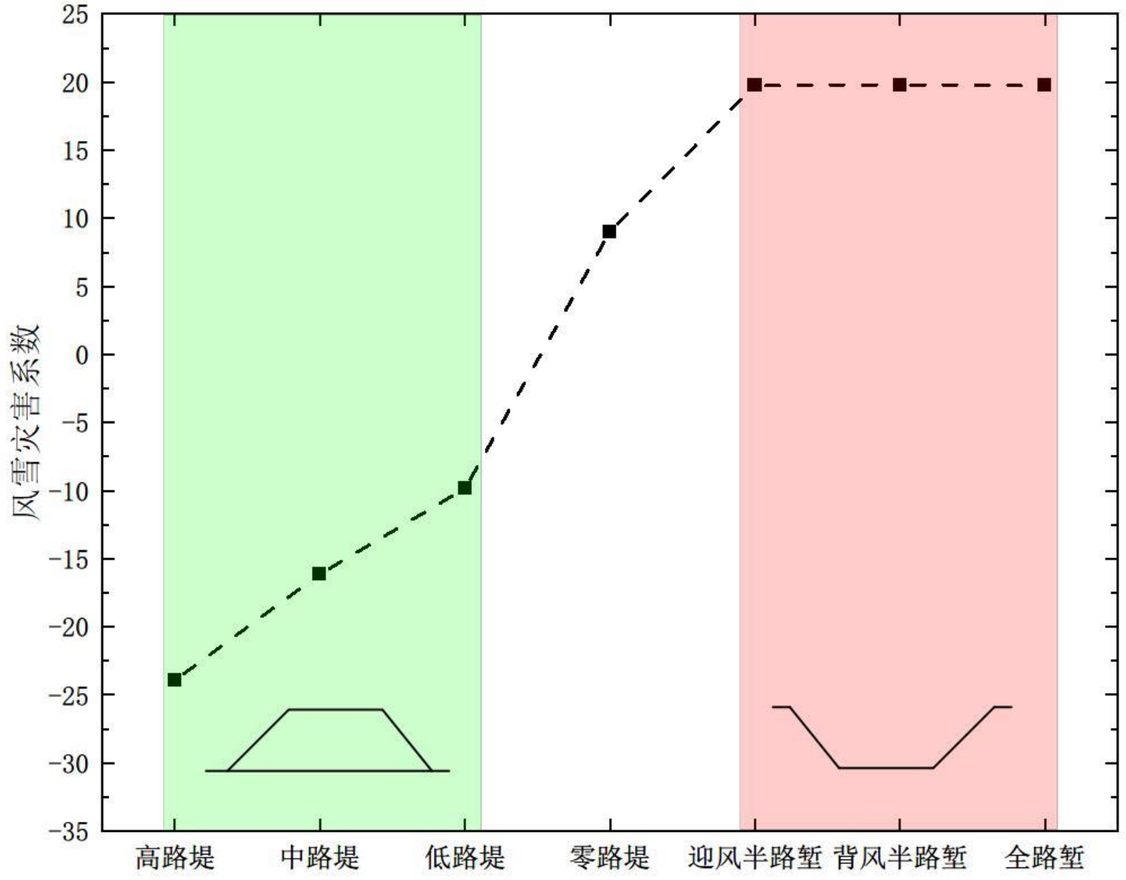
003 近年来交通线路风雪灾害频出

极寒天气造成新疆阿勒泰市7名工人遇难

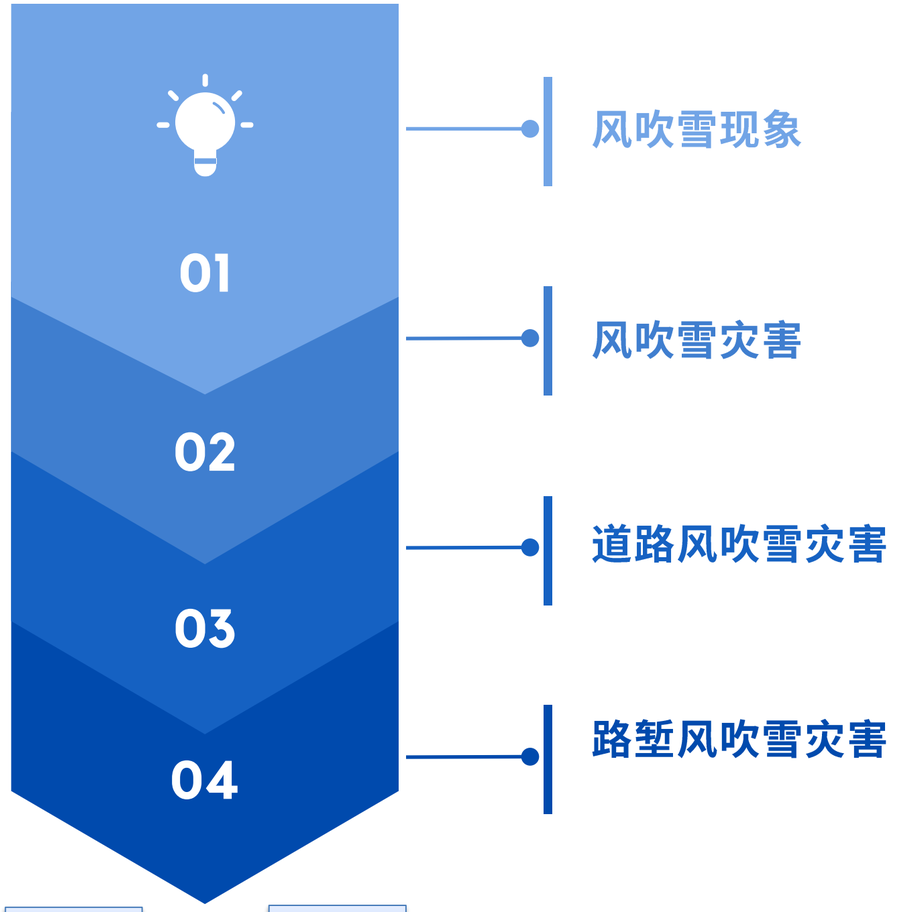
2022 11/29 08:14

新京报 | 热点 |

/ 路 堑 风 吹 雪 灾 害 /



* Li P, Bai M, Ding L, et al. Field measurement and outdoor wind tunnel test of snow-drifting: Snow distribution characteristics of railway subgrade and deposition mechanism[J]. Journal of Wind Engineering and Industrial Aerodynamics, 2022, 230: 105197.

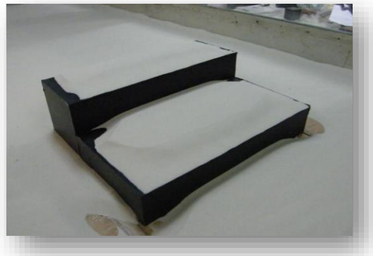


/ 风 雪 两 相 流 模 拟 /



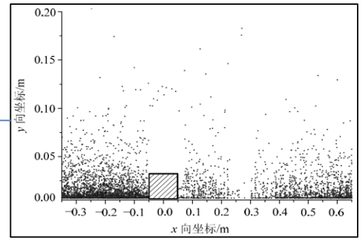
- **现场实测**

对足尺/缩尺模型开展实际户外观测；
数据真实可靠；
测量环境随机性大，受自然条件约束；



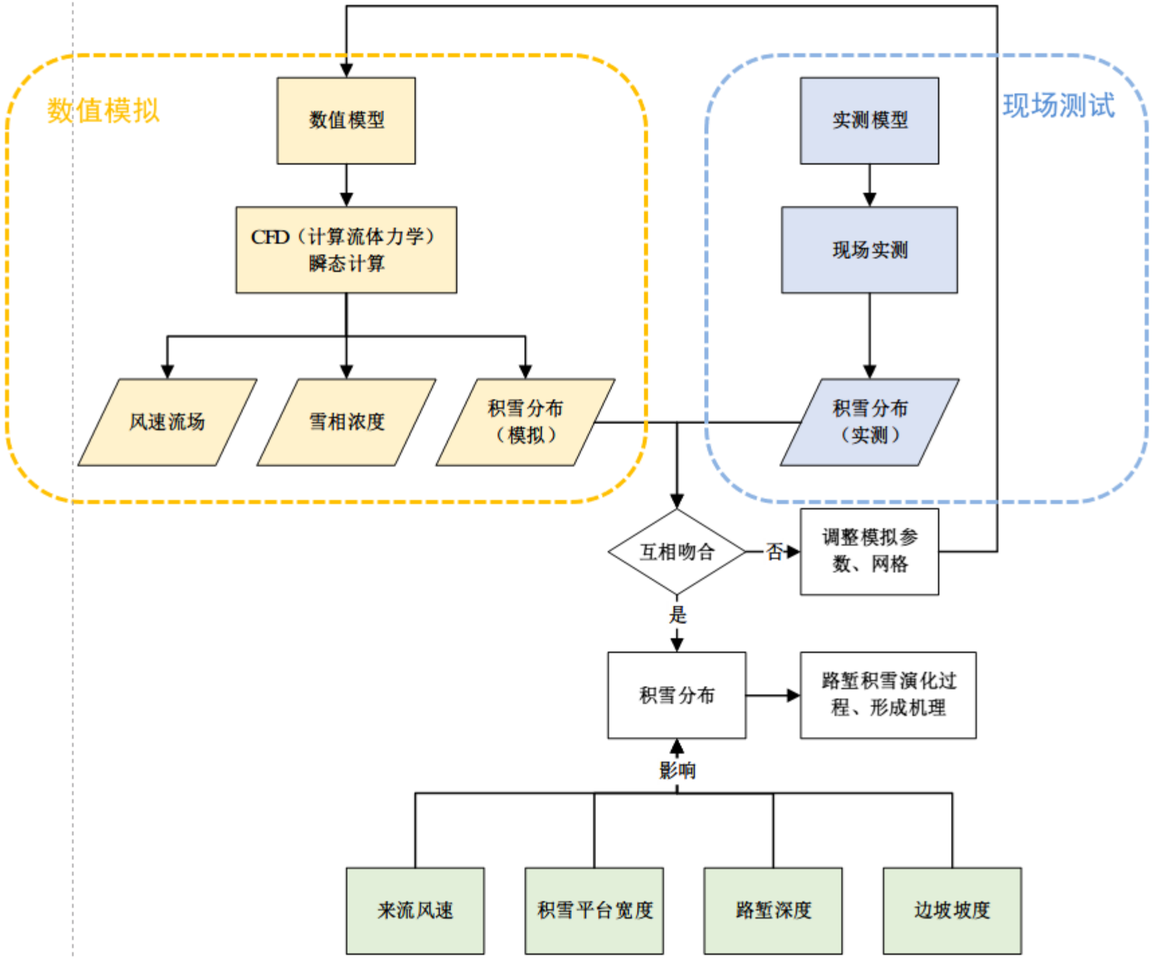
- **风洞试验**

借助人工设备，按照一定相似比例研究；
可重复性好，具有可探讨机理性；
对模拟物和试验条件要求高；

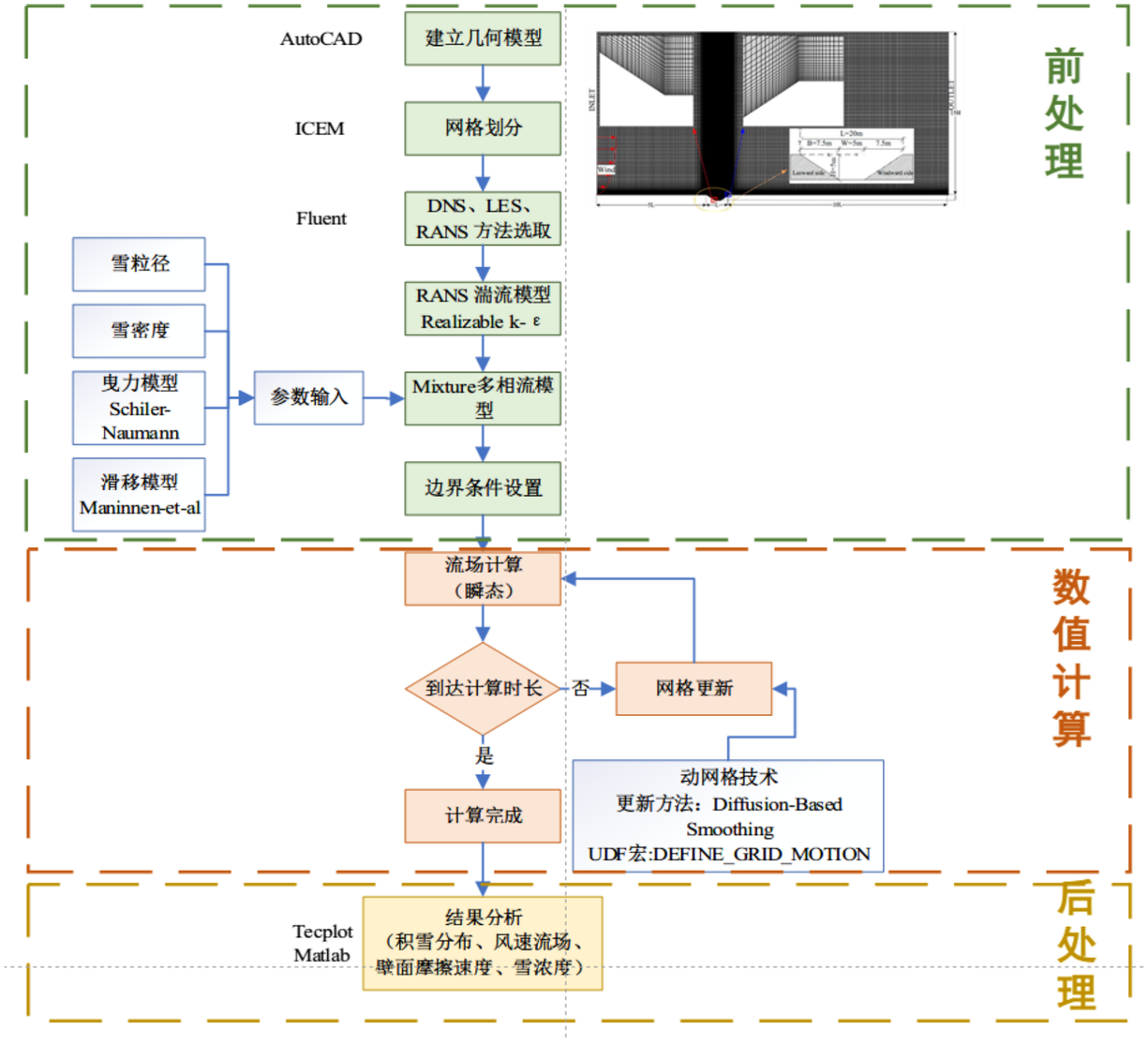


- **数值模拟**

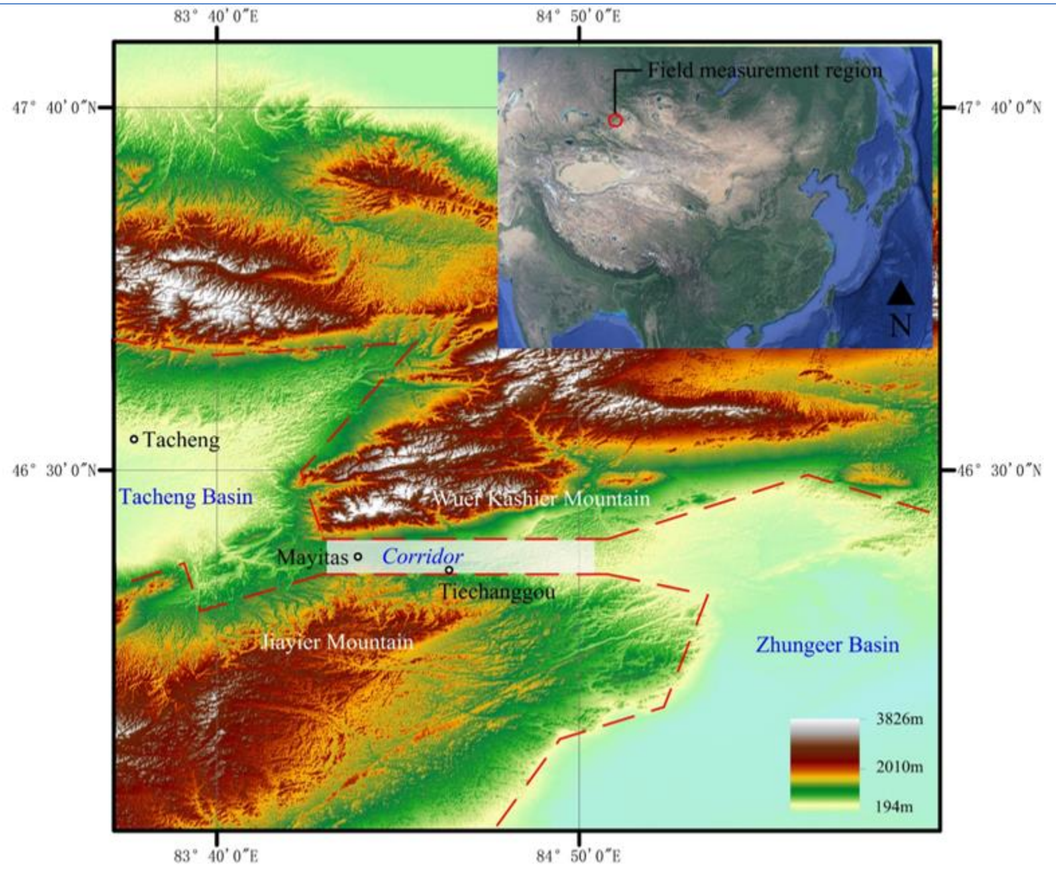
求解空气相与雪相的控制方程；
方便地研究各种参数对结果的影响；
依赖试验和实测的结果进行验证；



/ 风 雪 两 相 流 模 拟 /


前处理
AutoCAD
建立几何模型
ICEM
网格划分
Fluent
DNS、LES、RANS 方法选取
RANS 湍流模型 Realizable k- ε
雪粒径
雪密度
曳力模型 Schiler-Naumann
滑移模型 Maninnen-et-al
参数输入
Mixture多相流模型
边界条件设置
数值计算
流场计算（瞬态）
到达计算时长
否
网格更新
是
计算完成
动网格技术
更新方法：Diffusion-Based Smoothing
UDF宏:DEFINE_GRID_MOTION
后处理
Tecplot Matlab
结果分析（积雪分布、风速流场、壁面摩擦速度、雪浓度）

/ 风 吹 雪 现 象 的 形 成 /

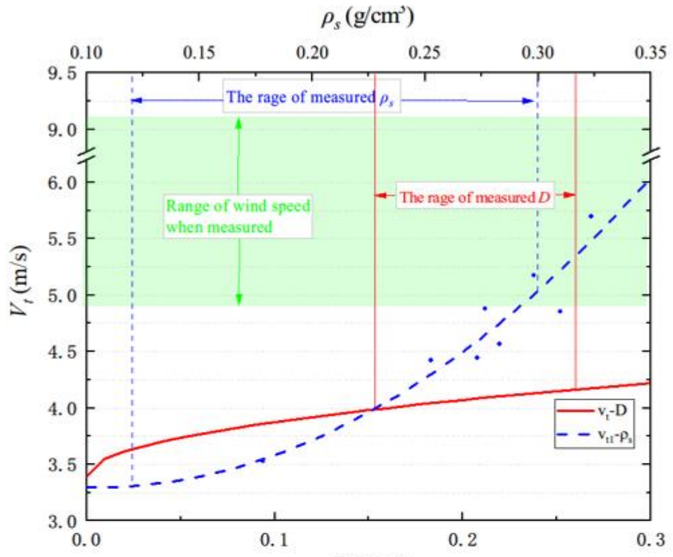


- **实测地点**

玛依塔斯地区，东西方向平坦，南北为山；

春冬两季有由西向东的西伯利亚寒流经过；

“狭道效应”显著，风力强劲；

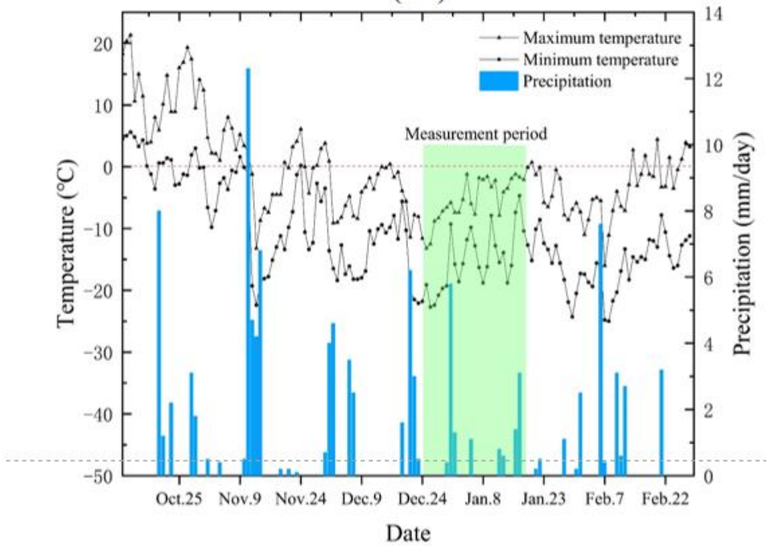


- **起动风速**

现场实测期间

实测风速始终大于起动风速

足够大的风速



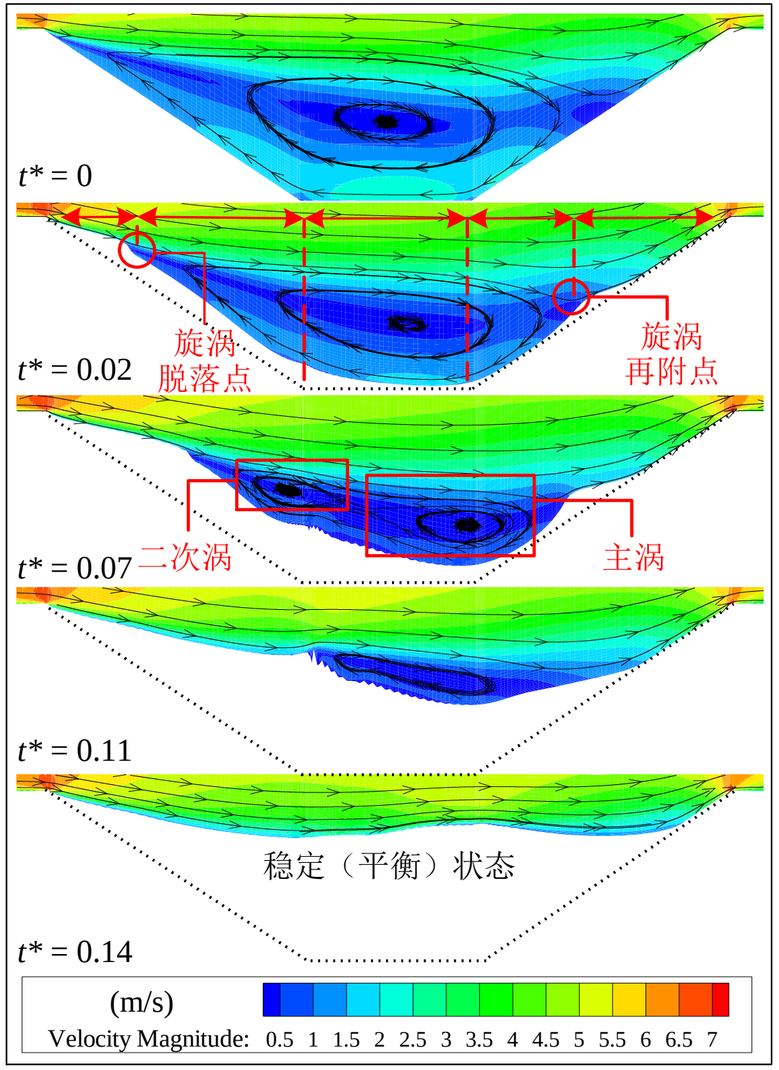
- **气温及降水**

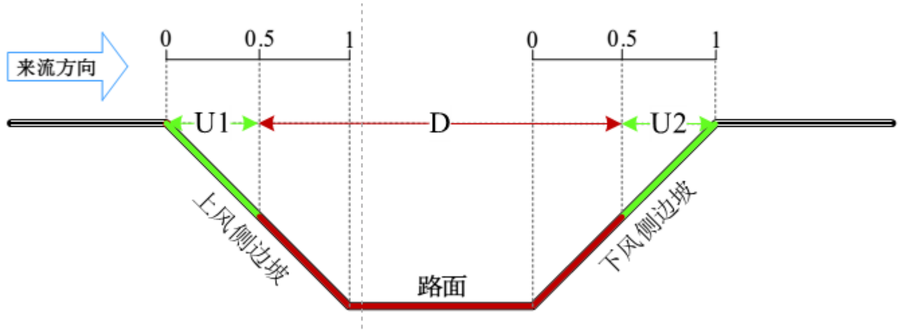
降雪次数较多；

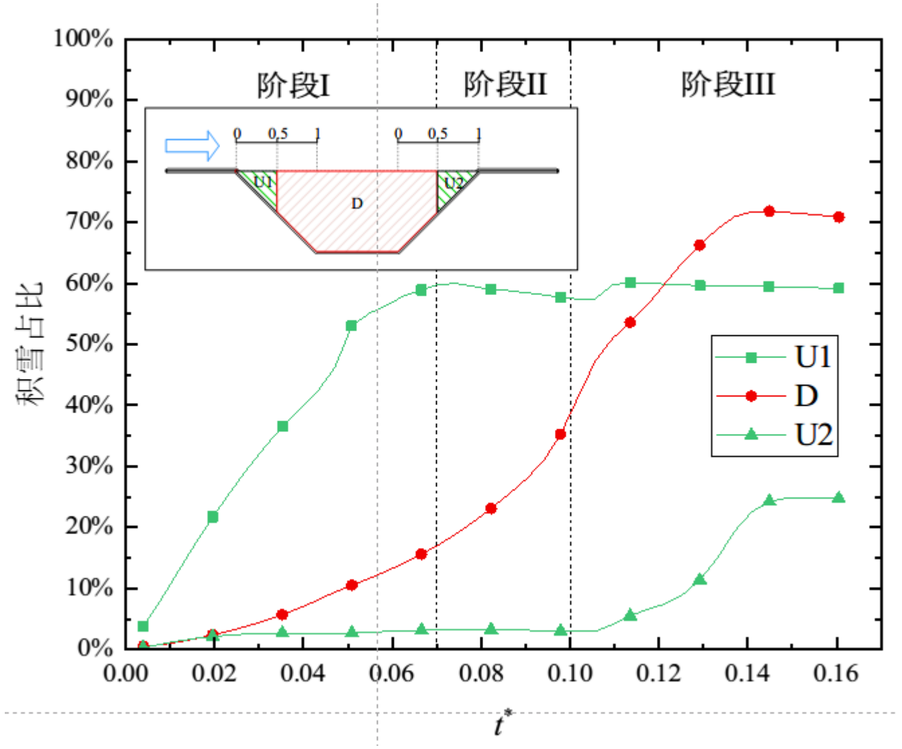
气温在0°C以下；

充足的、易起动（松散）的雪

/ 路 堑 风 吹 雪 灾 害 的 形 成 /

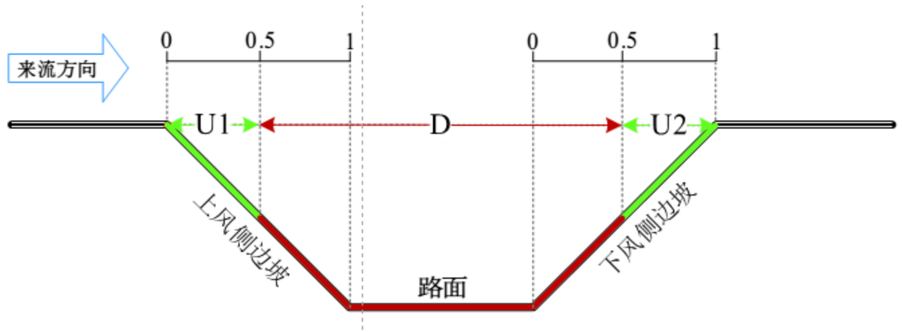




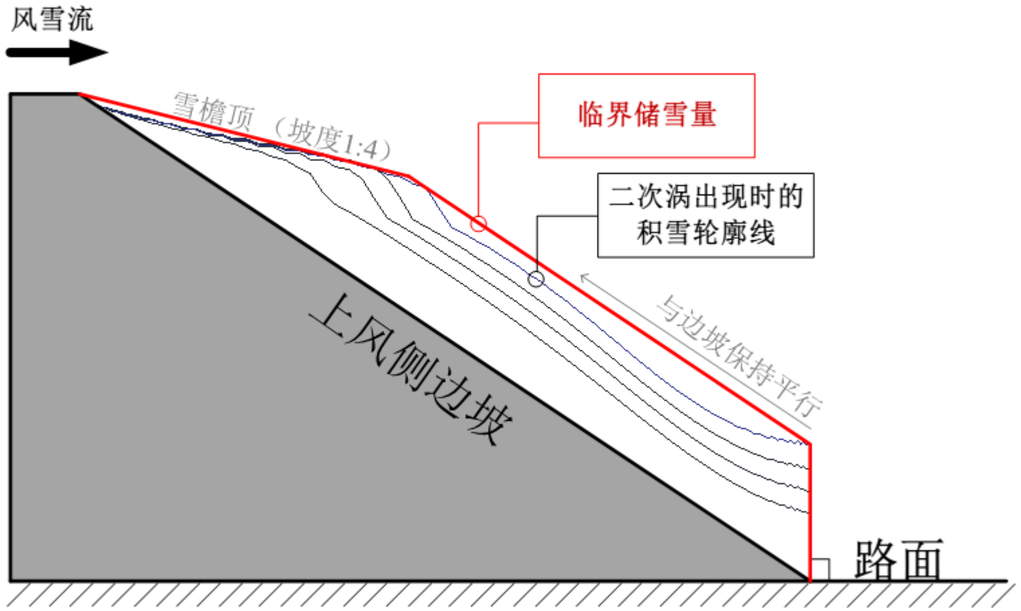


- 路堑内存在有一定尺度的旋涡，旋涡中心风速最小；上下侧边坡上分别有旋涡脱离点和旋涡再附点，积雪在两点之间的范围内增加；
- 积雪增加到一定程度，会有二次涡的出现，二次涡出现后路面积雪快速增加，当路面积雪足够多后，二次涡消失

/路堑风吹雪灾害的预测与防治/

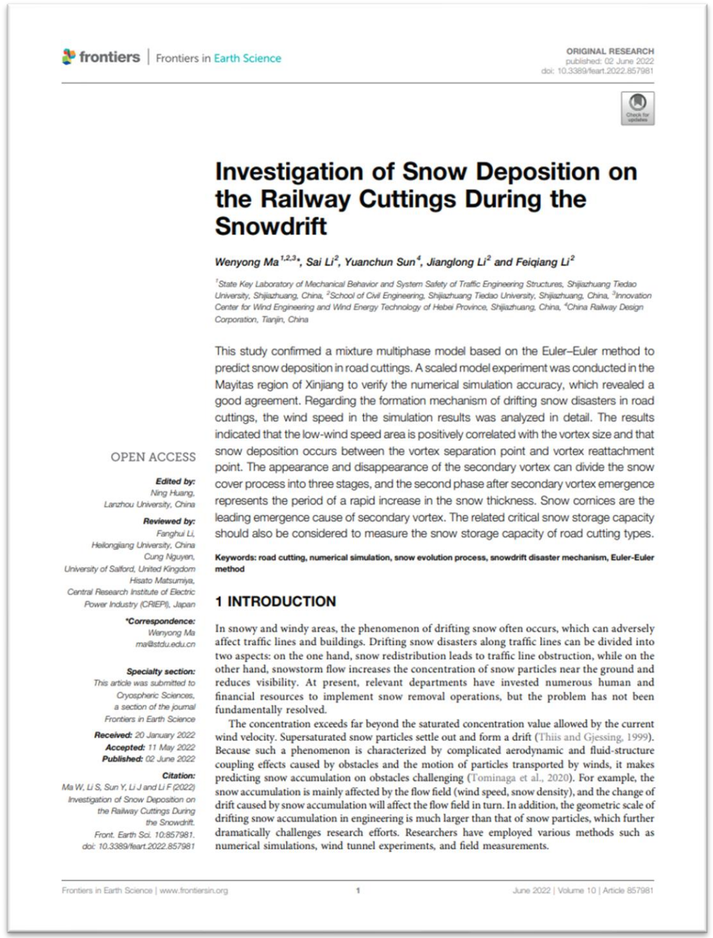


	积雪量变化趋势		
分区	阶段I	阶段II	阶段III
U1	快速增加	基本不变	基本不变
D	缓慢增加	快速增加	快速增加
U2	基本不变	基本不变	快速增加



预测供雪量 > 临界储雪量 ⟶ 风吹雪灾害

[期刊论文] Ma, Wenyong, **Sai Li**, Yuanchun Sun, Jianglong Li, and Feiqiang Li. "Investigation of Snow Deposition on the Railway Cuttings During the Snowdrift." *Frontiers in Earth Science* 10 (2022).



frontiers | Frontiers in Earth Science

ORIGINAL RESEARCH
published: 02 June 2022
doi: 10.3389/feart.2022.857981

Investigation of Snow Deposition on the Railway Cuttings During the Snowdrift

Wenyong Ma[1,2,3]*, Sai Li*[2]*, Yuanchun Sun*[4]*, Jianglong Li*[2] *and Feiqiang Li*[2]

[1]State Key Laboratory of Mechanical Behavior and System Safety of Traffic Engineering Structures, Shijiazhuang Tiedao University, Shijiazhuang, China, [2]School of Civil Engineering, Shijiazhuang Tiedao University, Shijiazhuang, China, [3]Innovation Center for Wind Engineering and Wind Energy Technology of Hebei Province, Shijiazhuang, China, [4]China Railway Design Corporation, Tianjin, China

This study confirmed a mixture multiphase model based on the Euler–Euler method to predict snow deposition in road cuttings. A scaled model experiment was conducted in the Mayitas region of Xinjiang to verify the numerical simulation accuracy, which revealed a good agreement. Regarding the formation mechanism of drifting snow disasters in road cuttings, the wind speed in the simulation results was analyzed in detail. The results indicated that the low-wind speed area is positively correlated with the vortex size and that snow deposition occurs between the vortex separation point and vortex reattachment point. The appearance and disappearance of the secondary vortex can divide the snow cover process into three stages, and the second phase after secondary vortex emergence represents the period of a rapid increase in the snow thickness. Snow cornices are the leading emergence cause of secondary vortex. The related critical snow storage capacity should also be considered to measure the snow storage capacity of road cutting types.

Keywords: road cutting, numerical simulation, snow evolution process, snowdrift disaster mechanism, Euler-Euler method

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1 INTRODUCTION

In snowy and windy areas, the phenomenon of drifting snow often occurs, which can adversely affect traffic lines and buildings. Drifting snow disasters along traffic lines can be divided into two aspects: on the one hand, snow redistribution leads to traffic line obstruction, while on the other hand, snowstorm flow increases the concentration of snow particles near the ground and reduces visibility. At present, relevant departments have invested numerous human and financial resources to implement snow removal operations, but the problem has not been fundamentally resolved.

The concentration exceeds far beyond the saturated concentration value allowed by the current wind velocity. Supersaturated snow particles settle out and form a drift (Thiis and Gjessing, 1999). Because such a phenomenon is characterized by complicated aerodynamic and fluid-structure coupling effects caused by obstacles and the motion of particles transported by winds, it makes predicting snow accumulation on obstacles challenging (Tominaga et al., 2020). For example, the snow accumulation is mainly affected by the flow field (wind speed, snow density), and the change of drift caused by snow accumulation will affect the flow field in turn. In addition, the geometric scale of drifting snow accumulation in engineering is much larger than that of snow particles, which further dramatically challenges research efforts. Researchers have employed various methods such as numerical simulations, wind tunnel experiments, and field measurements.

[期刊论文] 马文勇，李江龙，孙元春，**李赛**，白建方，风致路堑积雪演化过程试验与模拟研究．振动与冲击，2023，42(9)：284-292



振 动 与 冲 击
第 42 卷第 9 期 JOURNAL OF VIBRATION AND SHOCK Vol. 42 No. 9 2023

风致路堑积雪演化过程试验与模拟研究

马文勇[1,2,3,4]，李江龙[1]，孙元春[5]，李 赛[1]，白建方[1]

(1. 石家庄铁道大学 土木工程学院，石家庄 050043；
2. 石家庄铁道大学 道路与铁道工程安全保障省部共建教育部重点实验室，石家庄 050043；
3. 石家庄铁道大学 省部共建交通工程结构力学行为与系统安全国家重点实验室，石家庄 050043；
4. 河北省风工程和风能利用工程技术创新中心，石家庄 050043；
5. 中国铁路设计集团有限公司，天津 300251)

摘 要：路堑是交通线路常见的路基断面形式，因断面构造特征较路堤更容易发生风致雪灾害。为了给实际道路断面形式设计和风吹雪灾害防治提供建议和参考，采用风洞试验与数值模拟方法，研究了 4 种不同开敞形式的路堑在不同风速下风致雪漂移堆积演化过程，分析了风速和路堑开敞形式对积雪分布和演化过程的影响，并基于路面积雪量和积雪深度评估了不同路堑抵御雪灾害能力。结果表明：当上风侧边坡坡度大于下风侧边坡坡度时，路堑路面积雪量最少，路面的积雪最薄；增大风速和增加持续时间会加剧路面积雪，是影响积雪分布的重要因素；改变路堑边坡对减少路面积雪量作用有限，需设置其他防治措施来减轻道路风致雪灾害。

关键词：风吹雪；路堑；风洞试验；数值模拟；雪堆积
中图分类号：U213.1+2 **文献标志码**：A DOI:10.13465/j.cnki.jvs.2023.09.033

Tests and simulation for evolution process of wind-induced cutting snow accumulation

MA Wenyong[1,2,3,4]，*LI Jianglong*[1]，*SUN Yuanchun*[5]，*LI Sai*[1]，*BAI Jianfang*[1]

(1. School of Civil Engineering, Shijiazhuang Tiedao University, Shijiazhuang 050043, China;
2. Key Laboratory of Road and Railway Engineering Safety, Ministry of Education, Shijiazhuang Tiedao University, Shijiazhuang 050043, China; 3. State Key Laboratory of Structural Mechanics Behavior and System Safety in Traffic Engineering, Shijiazhuang Tiedao University, Shijiazhuang 050043, China;
4. Wind Engineering and Wind Energy Utilization Engineering Technology Innovation Center of Hebei Province, Shijiazhuang 050043, China; 5. China Railway Design Corporation, Tianjin 300251, China)

Abstract: Cutting is a common form of subgrade section for traffic lines. Due to its section structural characteristics, it is more likely to have wind-induced snow disasters than embankment. Here, to provide suggestions and reference for design of actual road section form and prevention and cure of wind-blown snow disasters, wind tunnel tests and numerical simulation methods were used to study evolution processes of wind-induced snow drift and accumulation with 4 different cutting open forms under different wind speeds, effects of wind speed and cutting open form on distribution and evolution process of snow cover were analyzed, and different cuttings' abilities to resist snow disasters were evaluated based on snow cover amount and depth on road surface. The results showed that when slop of windward side is larger than slope of leeward side, snow amount on cutting surface is the least, and snow thickness on road surface is the thinnest; increasing wind speed and duration can aggravate snow cover on road surface, so it is an important factor affecting distribution of snow cover; changing cutting slope has limited effect on reducing snow amount on road surface, and other prevention and cure measures need to be taken to reduce wind-induced snow disasters.

Key words: snow drift; road cutting; wind tunnel test; numerical simulation; snow accumulation

路堑是一种广泛应用在山区交通线路上的路基断面形式且相较于路堤更易发生风吹雪灾害[1]。当发生风吹雪现象时，携带雪颗粒的气流流经路堑时其运动速度迅速降低，同时流场结构发生显著变化，致使漂移雪粒在路面发生沉积，路面上空雪颗粒浓度急剧升高，

基金项目：河北省自然科学基金(E2021210053)
收稿日期：2022-04-12 修改稿收到日期：2022-05-23

[发明专利] CN202211180739.6，马文勇，**李赛**，王贺朋，陈伟，一种柔性光伏支架的索力调节装置和方法



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李胜强(010-82873665)

发文日：
2022 年 12 月 29 日

申请号或专利号：**202211180739.6** 发文序号：**2022122600442210**

申请人或专利权人： 石家庄铁道大学

发明创造名称： 一种柔性光伏支架的索力调节装置和方法

发 明 专 利 申 请 公 布 及 进 入 实 质 审 查 阶 段 通 知 书

上述专利申请，经初步审查，符合专利法实施细则第 44 条的规定。根据专利法第 34 条的规定，该申请在 38 卷 5102 期 2022 年 12 月 23 日专利公报上予以公布。

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210308
2018.10

纸件申请，回函请寄：100088 北京市海淀区蓟门桥西土城路 6 号 国家知识产权局专利局受理处收
电子申请，应当通过电子专利申请系统以电子文件形式提交相关文件。除另有规定外，以纸件等其他形式提交的文件视为未提交。



谢谢观看

恳请各位专家和老师批评指正

汇报人：李　赛

2023/4/28